Aufgabe 1:

Gesucht ist die kleinste Teilmenge χ der Potenzmenge von $\{0,1,2,3,4\}$, welche folgende Eigenschaften besitzt:

1. $\{0,2,4\} \in \chi$
2. $A \in \chi \implies \{0,1,2,3,4\} \setminus A \in \chi$
3. $A, B \in \chi \implies A \cup B \in \chi$

$\{0,2,4\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \{0,2,4\} = \{1,3\} \in \chi$

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$\overset{3.}{\implies} \{0,2,4\} \cup \{1,3\} = \{1,3\} \cup \{0,2,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \{0,1,2,3,4\} = \emptyset \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \{0,2,4\} = \{0,2,4\} \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \{1,3\} = \{1,3\} \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \emptyset = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \{0,2,4\} \cup \emptyset = \emptyset \cup \{0,2,4\} = \{0,2,4\} \in \chi$

$\overset{3.}{\implies} \{1,3\} \cup \emptyset = \emptyset \cup \{1,3\} = \{1,3\} \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \emptyset = \emptyset \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \emptyset \cup \emptyset = \emptyset \in \chi$

Endergebnis: $\chi = \{\emptyset, \{0,2,4\}, \{1,3\}, \{0,1,2,3,4\}\}$

Aufgabe 2:

$x = (5,4,3,2,1) \wedge y = (16,9,4,1,0)$

$\overline{x} = \dfrac{5+4+3+2+1}{5} = \dfrac{15}{5} = 3$

$\overline{y} = \dfrac{16+9+4+1+0}{5} = \dfrac{30}{5} = 6$

$x - \overline{x} = (5-3, 4-3, 3-3, 2-3, 1-3) = (2,1,0,-1,-2)$

$y - \overline{y} = (16-6, 9-6, 4-6, 1-6, 0-6) = (10,3,-2,-5,-6)$

$s(x) = \sqrt{\dfrac{1}{5-1}\sum_i (x_i - \overline{x})^2} = \sqrt{\dfrac{1}{4}(2^2 + 1^2 + 0^2 + (-1)^2 + (-2)^2)}$

$= \sqrt{\dfrac{(4+1+0+1+4)}{4}} = \sqrt{\dfrac{10}{4}} = \dfrac{\sqrt{10}}{2}$

$s(y) = \sqrt{\dfrac{1}{5-1}(\sum_i (y_i - \overline{y})^2)} = \sqrt{\dfrac{1}{4}(10^2 + 3^2 + (-2)^2 + (-5)^2 + (-6)^2)}$

$= \sqrt{\dfrac{(100+9+4+25+36)}{4}} = \sqrt{\dfrac{174}{4}} = \dfrac{\sqrt{174}}{2}$

$s(x,y) = \frac{1}{5-1}\sum_i (x_i - \overline{x})(y_i - \overline{y}) = \frac{1}{4}(2 \cdot 10 + 1 \cdot 3 + 0 \cdot -2 + (-1) \cdot (-5) + (-2) \cdot (-6))$

$= \dfrac{1}{4}(20 + 3 + 0 + 5 + 12) = \dfrac{1}{4} \cdot 40 = 10$

$r(x,y) = \dfrac{s(x,y)}{s(x)s(y)} = \dfrac{10}{\frac{\sqrt{10}}{2} \,.\, \frac{\sqrt{174}}{2}} = \dfrac{10}{\frac{\sqrt{1740}}{4}} = \dfrac{40}{\sqrt{1740}} \approx 0,9589$

(a)

Die Ausgabe ist folgendes Bild:

